

$$Z_{MRB}^B = \text{SiLU}(\text{Linear}(\tilde{X}_{Emb})), \quad Z_{MRB}^M = \text{SSM}(\text{SiLU}(\text{Conv}(\text{Linear}(\tilde{X}_{Emb})))), \quad Z_{MRB} = \text{Linear}(Z_{MRB}^M \odot Z_{MRB}^B) + \tilde{X}_{Emb}$$

$$h_t = \overline{A}h_{t-1} + \overline{B}\tilde{x}_t, \quad \overline{A} = DR_A(\Delta, A) = \exp(\Delta A), \quad \overline{B} = DR_B(\Delta, A, B) = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$$